

Gurugram University
Scheme of Studies and Examination
Bachelor of Technology (Civil Engineering)
Semester-III

Sr. No.	Course Code	Course Title	Hours per week	Contact hours per week	Credit	Examination Schedule (Marks)				Duration of Exam (Hours)
			L-T-P			Class work	Theory	Practical	Total	
1.		Economics For Engineers	3-0-0	3	3	30	70	-	100	3
2.		Building construction and materials	3-0-0	3	3	30	70	-	100	3
3.		Mathematics III	3-0-0	3	3	30	70	-	100	3
4.		Engineering Mechanics	3-0-0	3	3	30	70	-	100	3
5.		Fluid Mechanics-I	3-0-0	3	3	30	70	-	100	3
6.		Surveying -I	3-0-0	3	3	30	70	-	100	3
7.		Building Drawing lab	0-0-2	1	1	50	-	50	100	3
8.		Engineering Mechanics Lab.	0-0-2	1	1	50	-	50	100	3
9.		Fluid Mechanics Lab.	0-0-2	1	1	50	-	50	100	3
10.		Surveying Lab.	0-0-2	1	1	50	-	50	100	3
TOTAL					22				1000	

Course Name	:	ECONOMICS FOR ENGINEERS	
Course Code	:		External marks: 70
Credits	:	3	Internal marks: 30
L-T-P	:	3-0-0	Total marks: 100

Course Objectives:

1. Acquaint the students to basic concepts of economics and their operational significance.
2. To stimulate the students to think systematically and objectively about contemporary economic problems

COURSE CONTENT

UNIT-1

Definition of Economics- Various definitions, types of economics- Micro and Macro Economics, nature of economic problem, Production Possibility Curve, Economic laws and their nature, Relationship between Science, Engineering, Technology and Economic Development.

Demand- Meaning of Demand, Law of Demand, **Elasticity of Demand-** meaning, factors effecting it, its practical application and importance.

UNIT-2

Production- Meaning of Production and factors of production, Law of variable proportions, Returns to scale, Internal and external economies and diseconomies of scale.

Various concepts of cost of production- Fixed cost, Variable cost, Money cost, Real cost, Accounting cost, Marginal cost, Opportunity cost. Shape of Average cost, Marginal cost, Total cost etc. in short run and long run.

UNIT-3

Market- Meaning of Market, Types of Market- Perfect Competition, Monopoly, Monopolistic Competition and Oligopoly (main features).

Supply- Supply and law of supply, Role of demand & supply in price determination and effect of changes in demand and supply on prices.

UNIT-4

Indian Economy- Nature and characteristics of Indian economy as under developed, developing and mixed economy (brief and elementary introduction), **Privatization** - meaning, merits and demerits.

Globalization of Indian economy - merits and demerits.

Banking- Concept of a Bank, Commercial Bank- functions, Central Bank- functions,

Difference between Commercial & Central Bank.

Course Outcomes: By the end of this course the student will be able to:

1. The students will able to understand the basic concept of economics.
2. The student will able to understand the concept of production and cost.
3. The student will able to understand the concept of market.
4. The student will able to understand the concept of privatization, globalization and banks.

Suggested Books:

1. Jain T.R., Economics for Engineers, VK Publication.
2. Chopra P. N., Principle of Economics, Kalyani Publishers.
3. Dewett K. K., Modern economic theory, S. Chand.
4. H. L. Ahuja., Modern economic theory, S. Chand.
5. Dutt Rudar & Sundhram K. P. M., Indian Economy.
6. Mishra S. K., Modern Micro Economics, Pragati Publications.
7. Singh Jaswinder, Managerial Economics, dreamtech press.
8. A Text Book of Economic Theory Stonier and Hague (Longman's Landon).
9. Micro Economic Theory – M.L. Jhingan (S.Chand).
10. Micro Economic Theory - H.L. Ahuja (S.Chand).
11. Modern Micro Economics : S.K. Mishra (Pragati Publications).
12. Economic Theory - A.B.N. Kulkarni & A.B. Kalkundrikar (R.Chand & Co).

BUILDING CONSTRUCTION AND MATERIALS			
Course Code		External marks:	70
Credits	3	Internal marks:	30
L-T-P	3-0-0	Total marks:	100
		Duration of Examination:	3 hrs

COURSE OBJECTIVES:

- To provide the students an overview of the profession of Civil Engineering.
- To give the students an illustration of the Civil Engineering, properties of various building material, basic requirements of a building and explain the building construction aspects.

COURSE CONTENT

UNIT-I

Module 1: Masonry Construction

Introduction, Various terms used in brick masonry, classification of bricks, composition, bonds in brick work, laying brick work, structural brick work, reinforced brick work, Defects in brick masonry, Stone masonry and its classification, composite masonry, Glass block masonry.

UNIT-II

Module 2: Timber, paints and varnishes

Classification of timber, structure of timber, seasoning of timber, defects in timber, fire proofing of timber, advantages of plywood and fiber boards, Important Indian timbers; Basic constituents of paints, types of paints, constituents of varnishes, characteristics and types of varnishes.

UNIT-III

Module 3: Roofs, Floors and Tiles

Types of roofs, various terms used, roof trusses-king post truss, queen post truss etc. Floor structures, ground, basement and upper floors, various types of floorings. Doors and Windows: Locations, sizes, types of doors and windows, fixtures and fasteners for doors and windows. **Tiles:** Manufacturing of tiles, Terra-cotta and its types, uses of terracotta.

Unit-IV

Module 4: Damp-Proofing, Water-Proofing and Fire protection

Dampness and its causes, prevention of dampness, materials used, damp-proofing treatment in buildings; Water proofing: water- proofing treatment of roofs; Fire protection: Fire resisting

construction, fire protection requirements for buildings. Sound insulation and Acoustics materials.

COURSE OUTCOMES:

At the end of the course, the students will be able to

- Explain the importance of Civil Engineering in the infrastructural development of the society.
- They will be able to illustrate the types and properties of various building materials.
- To be aware of various traditional building materials and also the emerging materials in the field of Civil Engineering construction.
- To select suitable type of flooring, Plastering, varnishes with their application.
- They should be able to describe the basic requirements to construct a building.

Suggested Books:

- Building Construction By Sushil Kumar, Standard Pub., N. Delhi
- Building Material By Rangawala
- Construction Engineering By Y.S. Sane
- Building Construction By Gurcharan Singh, Standard Pub., N. Delhi

Course Name	:	Mathematics III	
Course Code	:		External marks: 70
Credits	:	4	Internal marks: 30
L-T-P	:	3-1-0	Total marks: 100
Course Objectives:			

Course Objectives:

At the end of this course, the student should be able to learn the behaviour of civil engineering determinate structures under static and moving loads by analytical/experimental techniques and software tools. The student should also be able to acquire the ability to interpret and evaluate experimental results.

COURSE CONTENT

Unit-I

Transform Calculus: Fourier Series and Fourier Transforms : Euler's formulae, conditions for a Fourier expansion, change of interval, Fourier expansion of odd and even functions, Fourier expansion of square wave, rectangular wave, saw-toothed wave, half and full rectified wave, half range sine and cosine series. Fourier integrals, Fourier transforms, Shifting theorem (both on time and frequency axes) Fourier transforms of derivatives, Fourier transforms of integrals, Convolution theorem, Fourier transform of Dirac-delta function.

Unit-II

Cauchy-Riemann equations, necessary and sufficient conditions for a function to be analytic, polar form of the Cauchy-Riemann equations. Harmonic functions, application to flow problems. Integration of complex functions. Cauchy- Integral theorem and formula. Power series, radius and circle of convergence, Taylor's Maclaurin's and Laurent's series. Zeros and singularities of complex functions, Residues. Evaluation of real integrals using residues (around unit and semi circle only). Probability Distributions and Hypothesis Testing: Conditional probability, Bayes theorem and its applications, expected value of a random variable. Properties and application of Binomial, Poisson and Normal distributions

Unit-III

Numerical Methods-I: Solution of Polynomial and Transcendental equations – Bisection method, Regula-Falsi method and Newton-Raphson method, Interpolation using Newton's forward and backward difference formulae, Newton's divided difference and Lagrange's formulae, Numerical integration, Trapezoidal rule and Simpson's 1/3rd and 3/8 rules

Unit-IV

Numerical Methods – II: Ordinary differential equations: Taylor's series, Euler and modified Euler's methods. Runge-Kutta method of fourth order for solving first and second order equations. Milne's and Adam's predictor-corrector methods. Partial differential equations: Finite difference solution two dimensional Laplace equation and Poisson equation, Implicit and explicit methods for one dimensional heat equation (Bender-Schmidt and Crank-

Nicholson methods), Finite difference explicit method for wave equation.

Course Outcomes: By the end of this course the student will be able to:

1. To solve field problems in engineering involving partial differential equations
2. To find roots of polynomial and transcendental equations using numerical methods and conduct numerical integration
3. To deal with the Laplace transform and its application
4. To classify algebraic structure of any mathematical problem.

Suggested Books:

1. Erwin Kreyszig, Advanced Engineering Mathematics, John Wiley & Sons
2. B. V. Ramana, Higher Engineering Mathematics, Tata McGraw-Hill Publishing Company Limited
3. B. S. Grewal, Higher Engineering Mathematics, Khanna Publishers
4. P. Kandasamy, K. Thilagavathy, K. Gunavathi, Numerical Methods, S. Chand and Company
5. S. S. Sastry, Introductory Methods of Numerical Analysis, PHI.
6. N. P. Bali and Manish Goyal, A text book of Engineering Mathematics, Laxmi Publications.
7. C. L. Liu, Elements of Discrete Mathematics, Tata McGraw-Hill.
8. K. H. Rosen, Discrete Mathematics and its Applications, Tata McGraw-Hill.
9. J. L. Hein, Discrete Structures, Logic and Computability, Jones and Bartlett.

ENGINEERING MECHANICS			
Course Code		External marks:	70
Credits	3	Internal marks:	30
L-T-P	3-0-0	Total marks:	100
		Duration of Examination:	3 hrs

COURSE OBJECTIVES:

- Students should be able to identify and analyse the basic structural elements.
- Students can apply the concepts of analysis for the design of various civil engineering structures.
- Covers the relationship between stress and strain on deformable solids, principal stresses, maximum shearing stress, and the stresses acting on a structural member.
- To provide basic knowledge in mechanics of materials so that the students can solve real engineering problems and design engineering systems.

COURSE CONTENT

UNIT-A

Module 1: Simple Stresses and Strains

Properties of Materials, i.e. tensile test, idealized stress- strain diagrams, isotropic, linear, elastic, Visco- elastic and plastic materials, Concept of stresses and strains, St.Venant's principle, relationship between elastic constants, Poisson's Ratio, Hoop stress, Stress and extension of uniform bar and tapered bar under its own weight and due to load, stresses produced in compound bars due to axial loads, Factor of Safety, Thermal stress and strain calculations, Shear stresses and shear strain, Complementary shear stress.

Module 2: Compound stress and strains

Normal stress, tangential Stresses, Stresses induced due to Uniaxial loads, stresses induced by state of simple shear, stresses induced due to biaxial loads, Mohr's Circle (Graphical Method), Principal stresses and principal planes, Maximum shear stresses, Proof stress.

UNIT-B

Module 3: Shear Force and Bending Moment in Beams and Frames

Type of loads, Shear force and bending moment, relation between Shear force and bending moment, Definition and Sign conventions, axial force, Shear force and Bending moment diagrams

Module 4: Bending stresses and Shear stresses in Beam

Pure bending, bending stresses, combined bending and direct stresses, Middle Third rule, composite beams, Variation of shear stresses for various cross-sections of a beam.

UNIT-C

Module 5: Torsion and Thin Cylinder

Torsion equation, its applications to the hollow and solid circular shafts, comparison of solid and hollow Shafts, shafts in series and parallel. Combined torsion and bending of circular shafts. Introduction to thin cylinder, Stresses in thin cylinder vessels subjected to internal pressure Circumferential stresses (Hoop Stresses), longitudinal stress.

Module 6: Column and Strut

Criteria for stability of columns, Buckling of columns, Euler's formula for various end restraints, Rankine's formula, eccentrically loaded struts, struts with initial curvature, struts with lateral loading.

UNIT-D**Module 7: Analysis of Plane Trusses**

Different types of trusses, Analysis of plane determinate trusses by method of joints, method of UNITs and analysis of Space Trusses using Tension Coefficient Method.

Module 8: Failure Theories

Theories of failure: maximum principal stress theory, maximum principal strain theory, maximum shear stress theory, maximum strain energy theory, distortion energy theory, comparison of the failure theories.

Course Outcomes:

At the end of the course, the students will be able to

- Identify different materials and their behaviour.
- Analyse various civil engineering structures under different loading conditions.
- Apply the principles of structural mechanics in design structural elements.
- Apply the concepts of failure theories for design of structures.

Suggested Books:

- Strength of Material by G.H. Ryder, MacMillan Publishers India Ltd.
- Mechanics of Materials by E.J. Hearn, Elsevier Publications.
- Mechanics of Materials by Punmia and Jain, Laxmi Publications (P) Ltd.
- Mechanics of Materials by R.C.Hibbeler, Pearson Higher Education.
- Strength of Materials by Timoshenko and Young,, East West Press.
- Mechanics of materials by V Gupta, Narosa publishing house.

Fluid Mechanics-I			
Course Code		External marks:	70
Credits	3	Internal marks:	30
L-T-P	2-1-0	Total marks:	100
		Duration of Examination:	3 hrs

COURSE OBJECTIVES:

- Introduce the concepts of fluid mechanics useful in Civil Engineering applications.
- To provide the students a first level exposure related to fluid statics, kinematics and dynamics.
- To provide the knowledge for measurement of pressure, computations of hydrostatic forces on structural components, concepts of Buoyancy and their applications in many engineering problems.
- Topics included in this course are aimed to prepare a student to build a good fundamental background useful in the application-intensive courses covering hydraulics, hydraulic machinery and hydrology in later semesters.

COURSE CONTENT

UNIT A

Module 1: Basic Concepts and Definitions

Distinction between a fluid and a solid; Density, Specific weight, Specific gravity, Kinematic and dynamic viscosity; variation of viscosity with temperature, Newton law of viscosity; vapour pressure, cavitations; surface tension, capillarity, Bulk modulus of elasticity, compressibility, types of fluids

UNIT B

Module 2: Fluid Statics

Fluid Pressure: Pressure density height relationship, pressure at a point, Pascal's law, gauge and absolute pressure, Pressure variation with temperature, density and altitude. Piezometer, U-Tube Manometer, Single Column Manometer, U-Tube Differential Manometer, pressure gauges,

Module 3: Hydrostatic pressure and force

Hydrostatic pressure and force: horizontal, vertical and inclined surfaces, centre of pressure. Buoyancy and stability of floating bodies, metacentric height

UNIT C

Module 4: Fluid Kinematics

Classification of fluid flow : steady and unsteady flow; uniform and non-uniform flow; laminar and turbulent flow; rotational and irrotational flow; compressible and incompressible flow; ideal and real fluid flow; one, two and three dimensional flows; rotation and circulation; Stream line, path line, streak line and stream tube; stream function, velocity

potential function. One-, two- and three -dimensional continuity equations in Cartesian coordinates

Module 5: Fluid Dynamics

Equations of motion - Euler's equation; Bernoulli's equation – derivation; Energy Principle; limitations of Bernoulli's equation, Practical applications of Bernoulli's equation: Venturimeter, Orifice meter and Pitot tube

UNIT D

Module 6: Boundary Layer Analysis

Assumption and concept of boundary layer theory. Boundary-layer thickness, displacement, momentum & energy thickness, laminar and Turbulent boundary layers on a flat plate; Laminar sub-layer, smooth and rough boundaries, Local and average friction coefficients Separation and Control.

Module 7: Dimensional Analysis and Hydraulic Similitude

Dimensional analysis, Buckingham theorem, important dimensionless numbers and their significance, geometric, kinematic and dynamic similarity, model studies, physical modelling, similar and distorted models.

COURSE OUTCOMES:

- Understand the broad principles of fluid statics, kinematics and dynamics
- Understand definitions of the basic terms used in fluid mechanics
- Understand classifications of fluid flow
- Be able to apply the continuity, momentum and energy principles
- Be able to apply dimensional analysis

SUGGESTED BOOKS:

- Hydraulic and Fluid Mechanic by P.N.Modi & S.M.Seth
- Introduction to Fluid Mechanics by Robert W.Fox & Alan T.McDonald 3 Fluid Mechanics Through Problems by R.J.Garde
- Engineering Fluid Mechanics by R.J.Garde & A.G.Mirajgaoker
- Fluid Mechanic and Hydraulic machines R.K. BANSAL

Surveying-I			
Course Code		External marks:	70
Credits	3	Internal marks:	30
L-T-P	2-1-0	Total marks:	100
		Duration of Examination:	3 hrs

COURSE OBJECTIVES:

- To understand the importance of surveying in Civil engineering.
- To study the basics of linear, angular and direction measurements using chain/tape, theodolite and compass.
- To study the method of determination of height of points using various levelling method and tacheometer.
- To study the significance of Plane Table surveying in preparation of map and setting of different types of curves.

COURSE CONTENT

UNIT A

Module 1: Basics of Surveying and Linear measurement

Definition, principles of surveying, objectives and classifications, Instruments used for measuring distance, chaining, errors in chaining, tape corrections and examples, concept of Geoids and reference spheroids.

Module 2: Compass Surveying

Types of compass- prismatic and surveyor's compass, Bearings and meridians, declination, local attraction, errors and adjustments, Methods of compass traversing, checks in traversing, adjustment of closed traverse and examples.

UNIT B

Module 3: Levelling

Terms used in levelling, types of levels and staff, principles of levelling, temporary adjustments of levels, reduction of levels and booking of staff readings, examples.

Module 4: Geodetic Trigonometric levelling

Height and distances- base of the object accessible and inaccessible, geodetical observation, correction due to curvature and refraction, axis signal correction, difference in elevation between two points.

UNIT C

Module 5: Plane Table Surveying

Plane table accessories, methods of plane table surveying, sources of error, advantages and disadvantages of plane table surveying; contouring and characteristics of contour lines, locating contours, interpolation of contours, contour maps.

Module 6: Contouring

Contouring and characteristics of contour lines, locating contours, Methods of contouring, interpolation of contours, Uses of contour.

UNIT D

Module 6: Theodolite Surveying

Theodolite, parts of theodolite, Temporary adjustment of Theodolite, measurement of horizontal and vertical angles by different methods, theodolite traversing, adjustments of closed traverse.

Module 8: Tachometric surveying

Principle of of tacheometric surveying, different instrument used in tacheometry, stadia and tangential method of tacheometry, tacheometric constants and their determinations, examples.

COURSE OUTCOMES:

At the end of the course, the students will be able to

- To carry out surveying in the field for various civil engineering projects, prepare a contour map and plan of the area.
- Taking accurate measurements with different surveying instruments.
- Adjustment of traverse, and understand the process of setting of different curves for road and railway designs.

SUGGESTED BOOKS:

- Surveying volume I and II: B C Punmia.
- Engineering Surveying (Sixth Edition): W. Schofield.
- Text Book of Surveying: C. Venkataramiah.
- Introduction to GPS: The Global Positioning System: Ahmed El-Rabbany.
- Various Online resources including NPTEL.

Building Drawing Lab.			
Course Code		External marks:	50
Credits	1	Internal marks:	50
L-T-P	0-0-2	Total marks:	100
		Duration of Examination:	3 hrs

COURSE OBJECTIVES:

- To understand the principles of planning and bylaws.
- To draw plan, elevation and section of bond in brick work, walls and foundations, load bearing and framed structures.
- To prepare detailed working drawing for different parts of a building.

COURSE CONTENT

LIST OF EXPERIMENTS

1. Cavity Wall.
2. Different Bonds in brick work.
3. Grillage foundation.
4. Preparation of building drawing mentioning its salient features including the following details: a) Ground floor plan b) Two sectional Elevations c) Front and Side Elevations
5. Plan and sectional Elevation of different Stair-Cases.
6. Plan and sectional Elevation of different Doors and Windows.
7. Plan and sectional Elevation of different Ventilators.
8. Plan and sectional Elevation of Floors.
9. Plan and sectional Elevation of different roofs.

Course Outcomes:

At the end of the course, the students will be able to

- Student's ability to perform basic sketching techniques will improve.
- Students will be able to draw orthographic projections and sections.
- Student's ability to use architectural and engineering scales will increase.
- To prepare drawings for doors, windows, floors etc.
- To use various Symbols, Conventions and Abbreviations for building drawing,
- Prepare detail planning for single and two storied residential building and public building.

ENGINEERING MECHANICS LAB			
Course Code		External marks:	50
Credits	1	Internal marks:	50
L-T-P	0-0-2	Total marks:	100
		Duration of Examination:	3 hrs

COURSE OBJECTIVES:

- Structural Analysis experiments help to understand, to know the practical behaviour of the physical structures like beams, roof truss etc.
- A proper structural analysis of these structures helps the students to solve the practical problems.
- Different structural apparatus like Simply Supported Beam, Curved Member of different shape, Pin Joint Truss are studied in the laboratory.

LIST OF EXPERIMENTS

1. To determine elastic properties of a beam.
2. Torsion of cylindrical rods (Shaft).
3. To determine and analyse deflection of curved beams.
4. Experimental and analytical study of behaviour of struts with various end conditions.
5. To determine deflection of trusses – Horizontal and vertical deflection of various joints of a pin jointed truss.
6. Experimental and analytical study of a 3bar pin jointed Truss.
7. Experimental and analytical study of an elastically coupled beam.
8. To plot stress- strain curve for mild steel – Demonstration.

COURSE OUTCOMES:

At the end of the course, the students will be able to

- To acquire the knowledge about stresses and strains.
- To get knowledge about loading systems, types of supports and beams and understand the behaviour of different structural system for different loading and deflection.
- Able to calculate the about forces, moments and deflections.
- To verify and compare different theoretical and experimental theorems.
- Analyse and assess the behaviour and serviceability of the structures using analytical/experimental methods.

Fluid Mechanics-I Lab.			
Course Code		External marks:	50
Credits	1	Internal marks:	50
L-T-P	0-0-2	Total marks:	100

COURSE OBJECTIVES:

- To understand the physical processes of fluid more closely.
- Various apparatus like, Verification of Bernoulli's theorem apparatus, venturimeter & Orifice meters, orifice & mouth piece apparatus Flow over notches apparatus, vortex flow apparatus etc helps to understand different process.

LIST OF EXPERIMENTS

1. Verification of Bernoulli's Theorem
2. Calibration of V notch
3. Calibration of Rectangular notch
4. Calibration of Trapezoidal notch
5. Study of Pressure Measuring Devices
6. Determination of Metacentric height
7. Hydrostatics Force on Flat Surfaces/Curved Surfaces
8. Venturimeter
9. Orifice meter
10. Determination of coefficient C_d , C_v , and C_c

COURSE OUTCOMES:

At the end of the course, the students will be able to

- Verification of Bernoulli's theorem.
- Calibration of different notches, venturimeter and orifice meter.
- Determination of different coefficient and their verification.
- Study the different property of fluid flow.

Surveying-I Lab			
Course Code		External marks:	50
Credits	1	Internal marks:	50
L-T-P	0-0-2	Total marks:	100
		Duration of Examination:	3 hrs

COURSE OBJECTIVES:

- To use of Chain for linear measurement and traversing.
- To use of different compass for determination of directions and for traversing.
- To use different levels and determine the reduced levels, elevation and depressions of ground.
- To prepare maps using plane table by applying different methods.

LIST OF EXPERIMENTS

1. Chain Traversing
2. Compass Traversing
3. Fly Levelling
4. Cross sectioning
5. Profile levelling
6. Plane Table surveying: Radiation and Intersection
7. Resection- 2 and 3-point problem with plane Table
8. Contouring and preparation contour map.
9. Use of tangent clinometer

COURSE OUTCOMES:

On completion of the course, the students will be able to:

- Use conventional surveying tools such as chain/tape, compass, plane table, levels in the field for various civil engineering applications.
- Enter observation in field book, adjusting and plotting a traverse.
- To calculate the earth work for cutting and filling.
- To prepare contour maps of a small area and its importance in Civil Engineering.